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(54) **CURRENT LIMIT STRATEGY FOR HOIST APPLICATION**

(71) Applicant: **Hamilton Sundstrand Corporation**, Charlotte, NC (US)

(72) Inventors: **Anil Prasad Josyula**, Andhra Pradesh (IN); **Ashish Vijay**, Rajasthan (IN); **Vipul Kohli**, Gujarat (IN); **David Frederick Brookes**, Birmingham (GB)

(73) Assignee: **HAMILTON SUNDSTRAND CORPORATION**, Charlotte, NC (US)

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(51) **Int. Cl.**

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H02P 21/00	(2016.01)
H02P 21/18	(2016.01)
H02P 21/22	(2016.01)

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(58) **Field of Classification Search**

CPC H02P 21/22; H02P 21/18; H02P 21/001
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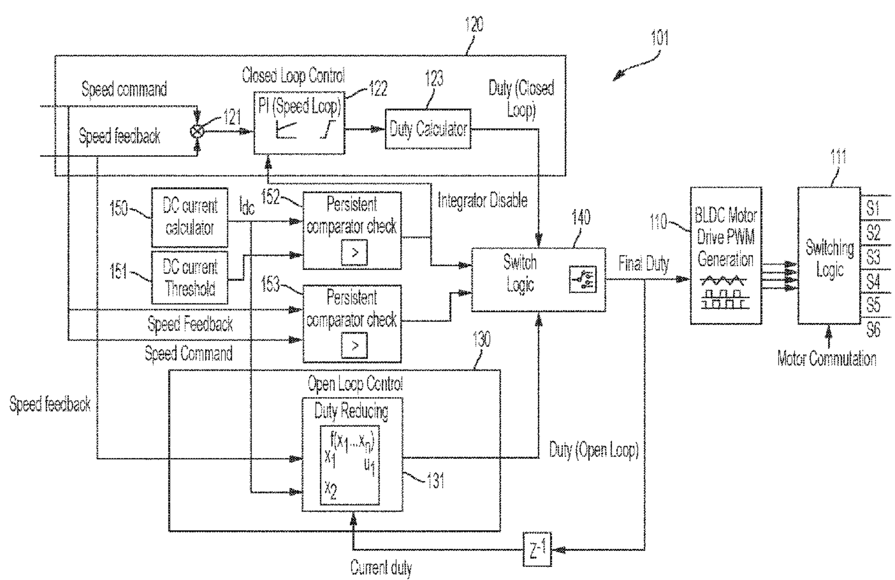
Primary Examiner — Erick D Glass
(74) *Attorney, Agent, or Firm* — CANTOR COLBURN LLP

(57) **ABSTRACT**

A hoist application system is provided. The hoist application system includes a motor drive unit, which is receptive of a final duty command and which drives a motor in accordance with the final duty command, a closed-loop control unit configured to generate a closed-loop duty command, an open-loop control unit configured to generate an open-loop duty command and a switch logic unit. The switch logic unit is receptive of direct current (DC) system information, the closed-loop duty command and the open-loop duty command, and is configured to generate the final duty command from one of the closed-loop duty command and the open-loop duty command based on the DC system information.

15 Claims, 4 Drawing Sheets



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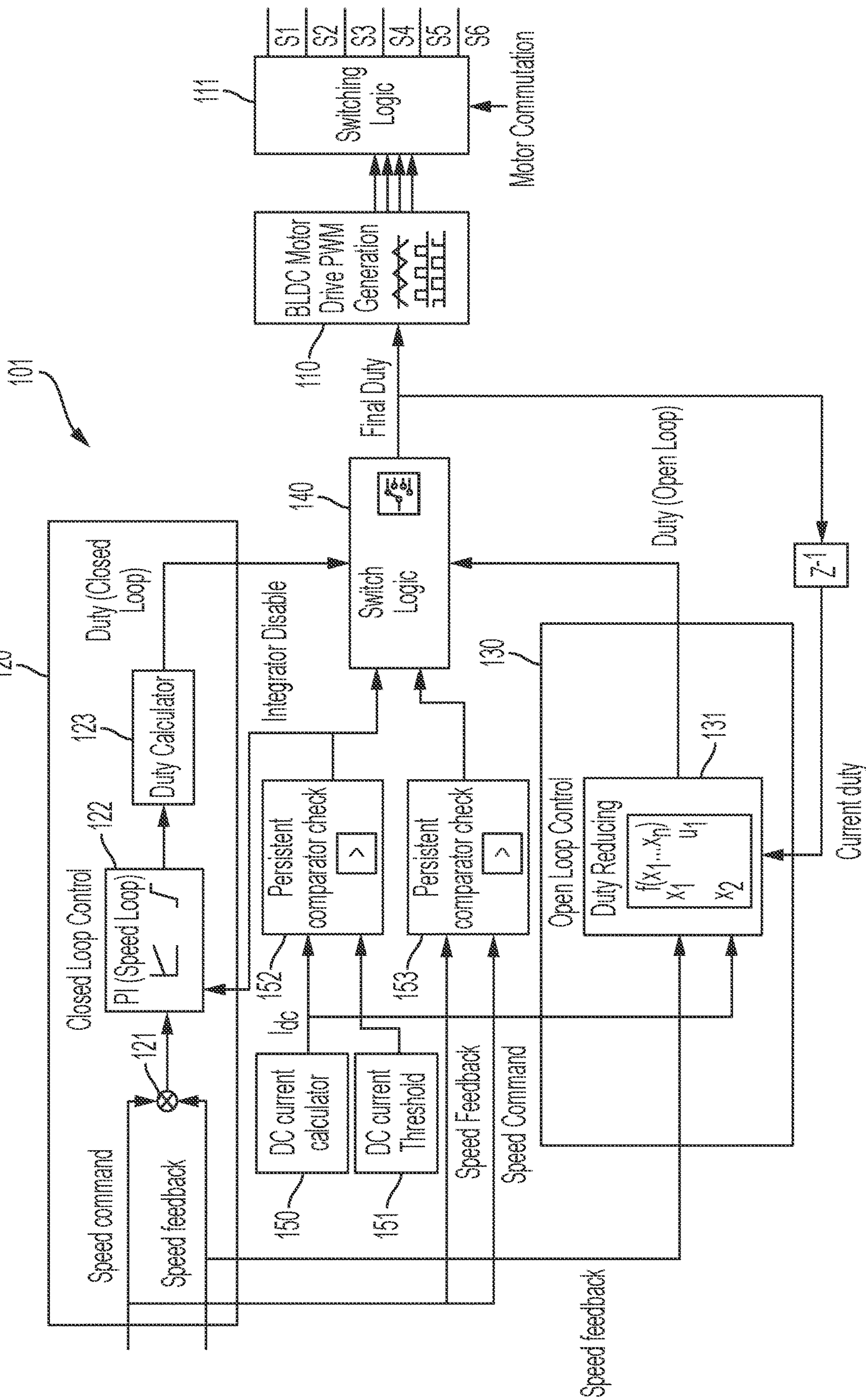
101
S1
S2
S3
S4
S5
S6
111
Switching Logic
Motor Commutation
BLDC Motor Drive PWM Generation
110
Final Duty
140
Switch Logic
Duty (Open Loop)
Z-1
Current duty
120
Closed Loop Control
Duty (Closed Loop)
123
Duty Calculator
122
PI (Speed Loop)
121
Speed command
Speed feedback
Integrator Disable
Persistent comparator check
>
152
Persistent comparator check
>
153
Idc
DC current calculator
150
DC current Threshold
151
Speed Feedback
Speed Command
130
Open Loop Control
131
Duty Reducing
$f(x_1 \ldots x_n)$
x_1
x_2
u_1
Speed feedback

FIG. 1

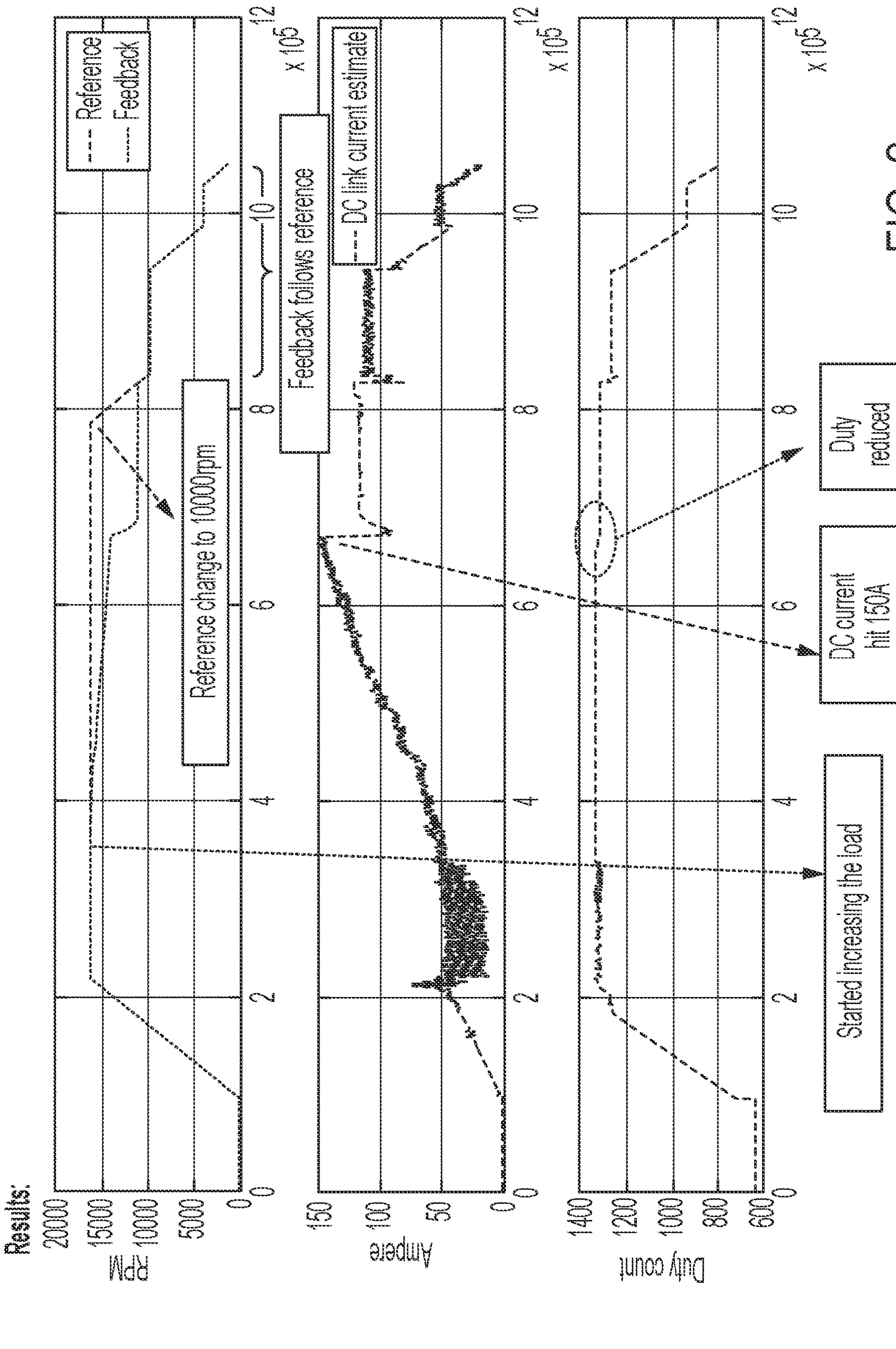
Results:
RPM
Ampere
Duty count
Reference
Feedback
DC link current estimate
x 10^5
Reference change to 10000rpm
Feedback follows reference
Started increasing the load
DC current hit 150A
Duty reduced

FIG. 2

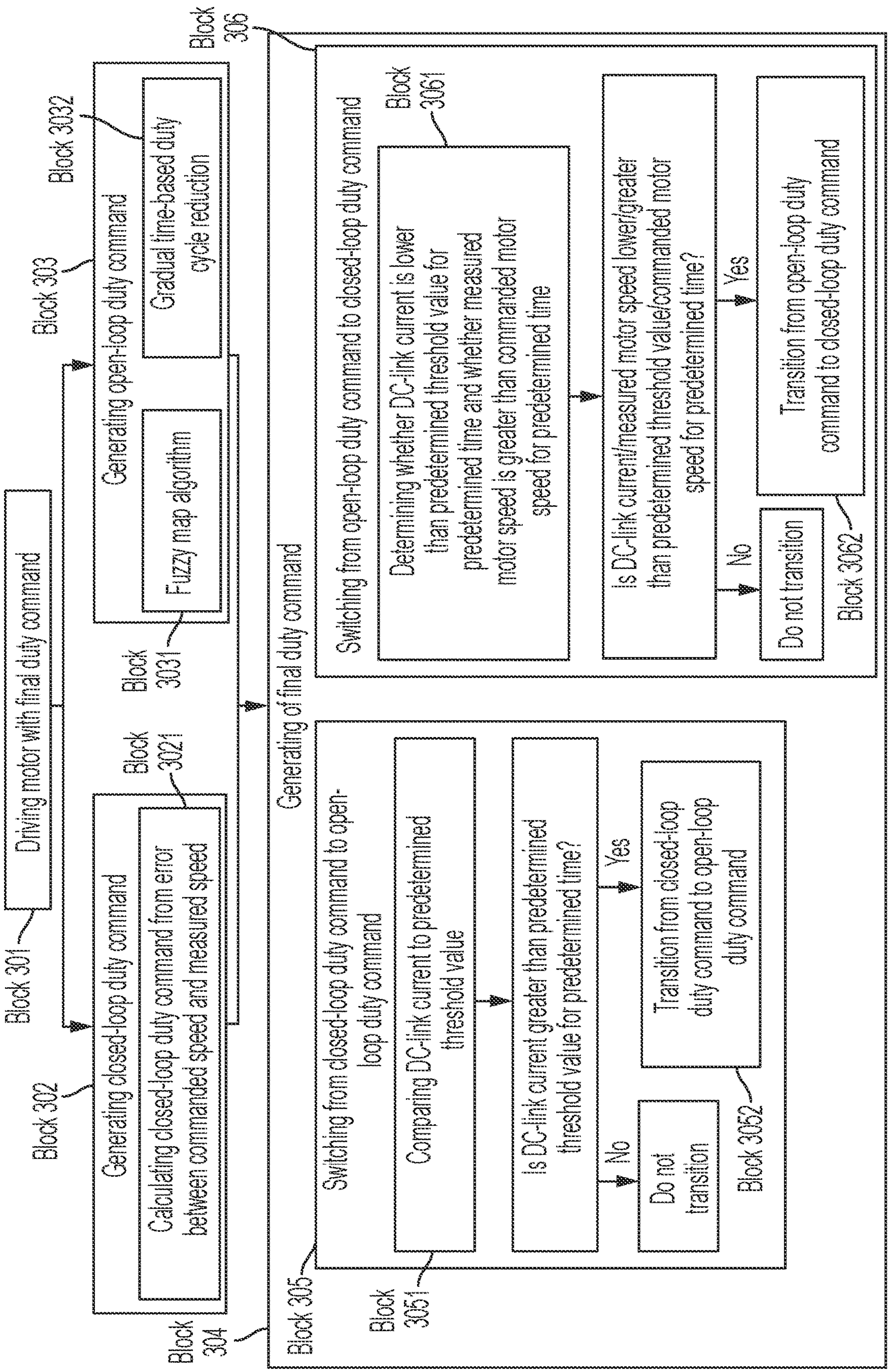
Block 301
Driving motor with final duty command
Block 302
Generating closed-loop duty command
Block 3021
Calculating closed-loop duty command from error between commanded speed and measured speed
Block 303
Generating open-loop duty command
Block 3031
Fuzzy map algorithm
Block 3032
Gradual time-based duty cycle reduction
Block 304
Generating of final duty command
Block 305
Switching from closed-loop duty command to open-loop duty command
Block 3051
Comparing DC-link current to predetermined threshold value
Is DC-link current greater than predetermined threshold value for predetermined time?
Yes
No
Do not transition
Block 3052
Transition from closed-loop duty command to open-loop duty command
Block 306
Switching from open-loop duty command to closed-loop duty command
Block 3061
Determining whether DC-link current is lower than predetermined threshold value for predetermined time and whether measured motor speed is greater than commanded motor speed for predetermined time
Is DC-link current/measured motor speed lower/greater than predetermined threshold value/commanded motor speed for predetermined time?
Yes
No
Do not transition
Block 3062
Transition from open-loop duty command to closed-loop duty command

FIG. 3

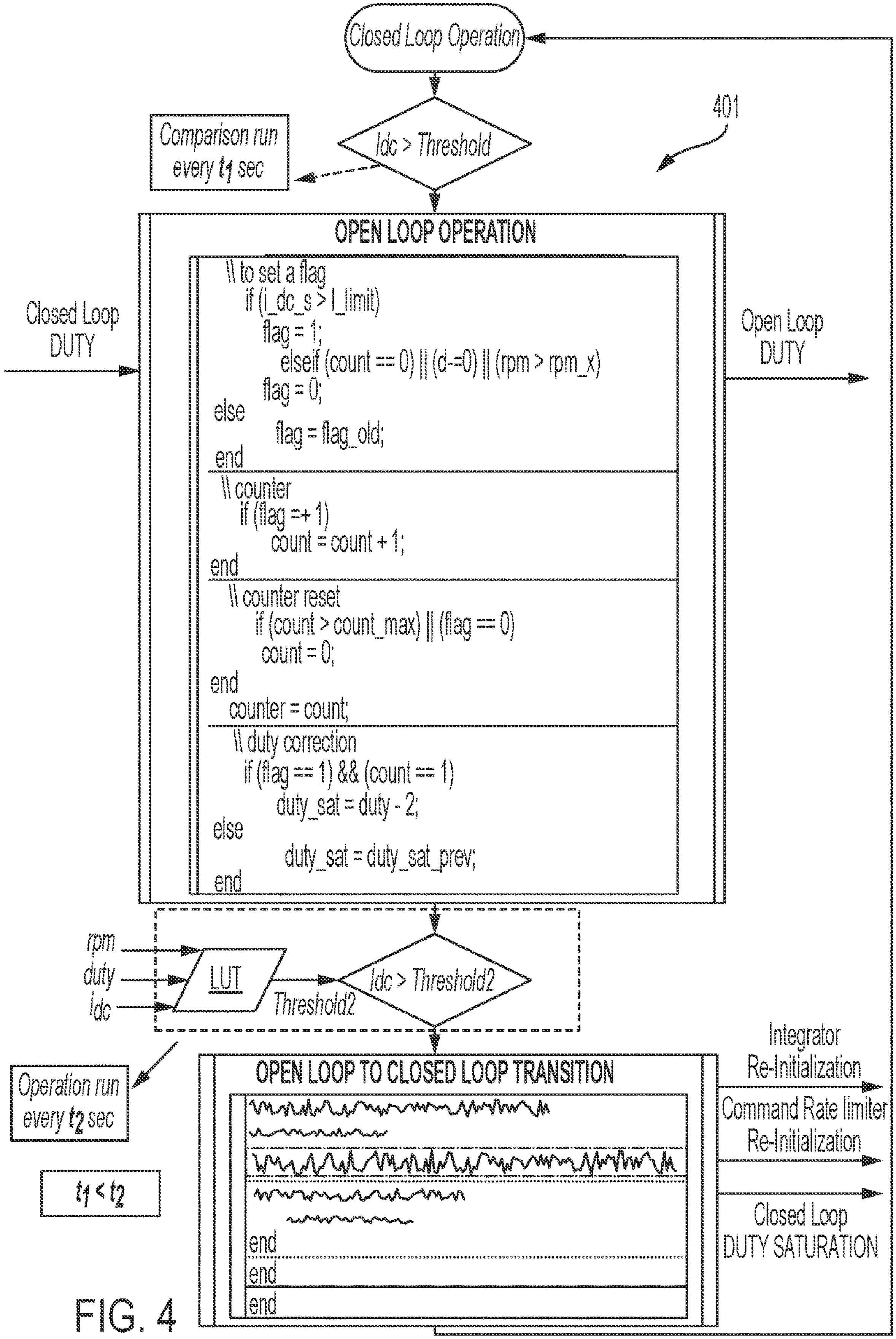
Closed Loop Operation
401
Idc > Threshold
Comparison run every t_1 sec
OPEN LOOP OPERATION
\\ to set a flag
if (i_dc_s > I_limit)
flag = 1;
elseif (count == 0) || (d-=0) || (rpm > rpm_x)
flag = 0;
else
flag = flag_old;
end
\\ counter
if (flag =+ 1)
count = count + 1;
end
\\ counter reset
if (count > count_max) || (flag == 0)
count = 0;
end
counter = count;
\\ duty correction
if (flag == 1) && (count == 1)
duty_sat = duty - 2;
else
duty_sat = duty_sat_prev;
end
Closed Loop DUTY
Open Loop DUTY
rpm
duty
Idc
LUT
Threshold2
Idc > Threshold2
Operation run every t_2 sec
$t_1 < t_2$
OPEN LOOP TO CLOSED LOOP TRANSITION
end
end
end
Integrator Re-Initialization
Command Rate limiter Re-Initialization
Closed Loop DUTY SATURATION

FIG. 4

CURRENT LIMIT STRATEGY FOR HOIST APPLICATION

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of Indian Application No. 202311020101 filed Mar. 22, 2023, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

The present disclosure relates to a hoist application and, in particular, to a current limit strategy for a hoist application.

Helicopter-hoist equipment typically includes a lifting device such as a hoist, which is attached to a helicopter, a hoist cable and a hook located at the distal end of the cable for direct or indirect attachment to a person, animal or object for rescue. The hoist usually has a rotary drum and a motor. The motor turns the rotary drum for winding the cable in and out to lift or transport a load.

A need remains for improved current control in a motor of helicopter-hoist equipment.

SUMMARY

According to an aspect of the disclosure, a hoist application system is provided. The hoist application system includes a motor drive unit, which is receptive of a final duty command and which drives a motor in accordance with the final duty command, a closed-loop control unit configured to generate a closed-loop duty command, an open-loop control unit configured to generate an open-loop duty command and a switch logic unit. The switch logic unit is receptive of direct current (DC) system information, the closed-loop duty command and the open-loop duty command, and is configured to generate the final duty command from one of the closed-loop duty command and the open-loop duty command based on the DC system information.

In accordance with additional or alternative embodiments, the closed-loop control unit is configured to calculate the closed-loop duty command based on an error between a commanded motor speed and a measured motor speed.

In accordance with additional or alternative embodiments, the open-loop control unit is configured to calculate the open-loop duty command using a fuzzy map algorithm.

In accordance with additional or alternative embodiments, the fuzzy map algorithm considers a measured motor speed, the DC system information and a current version of the final duty command.

In accordance with additional or alternative embodiments, the open-loop control unit is configured to calculate the open-loop duty command to achieve a gradual time-based duty cycle reduction.

In accordance with additional or alternative embodiments, the switch logic unit is configured to transition from the closed-loop duty command to the open-loop duty command by comparing a DC-link current to a threshold value from the DC system information and transitioning from the closed-loop duty command to the open-loop duty command in an event the DC-link current exceeds the threshold value for a predetermine time.

In accordance with additional or alternative embodiments, the threshold value is about 150 A.

In accordance with additional or alternative embodiments, the switch logic unit is configured to transition from the

open-loop duty command to the closed-loop duty command in an event the DC system information indicates that a DC-link current is lower than a threshold value for a predetermined time and closed-loop operation is possible without crossing the threshold value.

In accordance with additional or alternative embodiments, the threshold value is about 150 A.

According to an aspect of the disclosure, a method of operating a hoist application system is provided and includes driving a motor in accordance with a final duty command, generating, in a closed-loop control unit, a closed-loop duty command, generating, in an open-loop control unit, an open-loop duty command and generating the final duty command from one of the closed-loop duty command and the open-loop duty command based on direct current (DC) system information.

In accordance with additional or alternative embodiments, the generating of the closed-loop duty command includes calculating the closed-loop duty command based on an error between a commanded motor speed and a measured motor speed.

In accordance with additional or alternative embodiments, the generating of the open-loop duty command includes calculating the open-loop duty command using a fuzzy map algorithm.

In accordance with additional or alternative embodiments, the fuzzy map algorithm considers a measured motor speed, the DC system information and a current version of the final duty command.

In accordance with additional or alternative embodiments, the generating of the open-loop duty command includes calculating the open-loop duty command to achieve a gradual time-based duty cycle reduction.

In accordance with additional or alternative embodiments, the generating of the final duty command includes switching from the closed-loop duty command to the open-loop duty command and the switching includes comparing a DC-link current to a threshold value from the DC system information and transitioning from the closed-loop duty command to the open-loop duty command in an event the DC-link current exceeds the threshold value for a predetermine time.

In accordance with additional or alternative embodiments, the threshold value is about 150 A.

In accordance with additional or alternative embodiments, the generating of the final duty command includes switching from the open-loop duty command to the closed-loop duty command and the switching includes determining whether the DC system information indicates that a DC-link current is lower than a threshold value for a predetermined time and a closed-loop operation is possible without crossing the threshold value and switching from the open-loop duty command to the closed-loop duty command in an event the DC system information indicates that the DC-link current is lower than the threshold value for the predetermined time and, for a commanded motor speed and a load applied on the system, the closed-loop operation is possible without crossing the threshold value.

In accordance with additional or alternative embodiments, the threshold value is about 150 A.

According to an aspect of the disclosure, a method of operating a hoist application system is provided and includes driving a motor in accordance with a final duty command, generating, in a closed-loop control unit, a closed-loop duty command, generating, in an open-loop control unit, an open-loop duty command and generating the final duty command from one of the closed-loop duty command and the open-loop duty command based on direct current (DC)

system information. The generating of the final duty command includes one of switching from the closed-loop duty command to the open-loop duty command and switching from the open-loop duty command to the closed-loop duty command.

In accordance with additional or alternative embodiments, the switching from the closed-loop duty command to the open-loop duty command includes comparing a DC-link current to a threshold value from the DC system information and transitioning from the closed-loop duty command to the open-loop duty command in an event the DC-link current exceeds the threshold value for a predetermine time and the switching from the open-loop duty command to the closed-loop duty command includes determining whether the DC system information indicates that a DC-link current is lower than a threshold value for a predetermined time and a closed-loop operation is possible without crossing the threshold value and switching from the open-loop duty command to the closed-loop duty command in an event the DC system information indicates that the DC-link current is lower than the threshold value for the predetermined time and, for a commanded motor speed and a load applied on the system, the closed-loop operation is possible without crossing the threshold value.

Additional features and advantages are realized through the techniques of the present disclosure. Other embodiments and aspects of the disclosure are described in detail herein and are considered a part of the claimed technical concept. For a better understanding of the disclosure with the advantages and the features, refer to the description and to the drawings.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

For a more complete understanding of this disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, wherein like reference numerals represent like parts:

FIG. **1** is a schematic diagram illustrating a hoist application system in accordance with embodiments;

FIG. **2** is a graphical illustrating of an execution of the hoist application system of FIG. **1** in accordance with embodiments;

FIG. **3** is a flow diagram illustrating a method of operating the hoist application system of FIG. **1** in accordance with embodiments; and

FIG. **4** is a state diagram illustrating a hoist application system operation in accordance with embodiments.

DETAILED DESCRIPTION

For specific applications of a brushless direct current (DC) or BLDC motor, such as in a hoist application for a helicopter, there is often a desire to control hoisting speed without a current loop. In these cases, a challenge exists in that monitoring and limiting DC-link current becomes necessary. Indeed, for certain BLDC motors with very low inductance, it can be difficult to control the input current. Moreover, in certain low-cost drive systems, there may not be three-phase current sensors available.

Thus, as will be described below, a hoist application system is provided in which the DC-Link current is monitored and limited by taking a penalty on the speed of the motor in case of high loads. Typically, a hysteresis controller for the DC-link current can be used, but this will result in speed oscillations if it is not implemented with a high accuracy sensor or with a reduced latency system. To avoid this issue, a logical architecture is configured to smoothly transition from closed-loop speed control to an open-loop duty cycle-based motor drive system and vice versa. In the open-loop system, a calculated duty cycle is applied to limit the DC-link current based on speed and current feedback.

With reference to FIG. **1**, a hoist application system **101** is provided and includes a motor drive unit **110**, which is receptive of a final duty command and which drives a motor via switching logic **111** in accordance with the final duty command. The motor can be a BLDC motor and the motor drive unit **110** can be provided as a pulse width modulation (PWM) generation unit. The hoist application system **101** further includes a closed-loop control unit **120** that is configured to generate a closed-loop duty command, an open-loop control unit **130** that is configured to generate an open-loop duty command and a switch logic unit **140**. The switch logic unit **140** is receptive of direct current (DC) system information, the closed-loop duty command from the closed-loop control unit **120** and the open-loop duty command from the open-loop control unit **130**. The switch logic unit **140** is configured to generate the final duty command from one of the closed-loop duty command and the open-loop duty command based on the DC system information.

In addition, the hoist application system **101** also includes a DC current calculator **150**, which is configured to calculate DC-link current and to output a signal accordingly, a DC current threshold unit **151** that stores a predetermined threshold value of a DC current limit (i.e., about 150 A) and a first comparator **152**. The first comparator **152** is configured to compare the DC-link current and the predetermined threshold value, which together form the DC system information, to determine whether the DC-link current is currently above or below the predetermined threshold value and to issue a signal to the switch logic unit **140** accordingly. The hoist application system **101** still further includes a second comparator **153**, which is receptive of a motor speed command signal (i.e., a commanded motor speed) and a motor speed feedback signal (i.e., a measured motor speed). The second comparator **153** is configured to compare the commanded motor speed and the measured motor speed, the determine which is larger and to issue a signal to the switch logic unit **140** accordingly.

The closed-loop control unit **120** includes a summation unit **121**, which is receptive of the motor speed command signal (i.e., the commanded motor speed) and the motor speed feedback signal (i.e., the measured motor speed), and which is configured to generate an error signal that is representative of the error between the commanded motor speed and the measured motor speed. The closed-loop control unit **120** further includes a proportional integral (PI) controller **122**, which is receptive of the error signal that it converts to a PI voltage command signal, and a duty calculator **123**. The duty calculator **123** is receptive of the PI voltage command signal and thus calculates the closed-loop duty command. Thus, the closed-loop duty command is calculated based at least partially on the error between a commanded motor speed and a measured motor speed. The PI controller **122** can be disabled by the switch logic unit **140** in a case in which the switch logic unit **140** generates the final duty command from the open-loop duty command.

The open-loop control unit **130** includes a duty reducing unit **131**, which is receptive of a current version of the final duty command, the motor speed feedback signal and the DC-link current. The open-loop control unit **130** is thus configured to calculate the open-loop duty command using a fuzzy map algorithm, which considers a measured motor speed from the motor speed feedback signal, the DC-link current from the DC system information and the current version of the final duty command. Alternatively, the open-loop control unit **130** can be configured to calculate the open-loop duty command to achieve a gradual time-based duty cycle reduction.

With the hoist application system **101** configured as described above, the switch logic unit **140** is configured to transition from the closed-loop duty command to the open-loop duty command by comparing or recognizing the comparison of the first comparator **152** of the DC-link current to the predetermined threshold value (i.e., about 150 A) and transitioning from the closed-loop duty command to the open-loop duty command in an event the DC-link current exceeds the predetermined threshold value for a predetermine time. Conversely, the switch logic unit **140** is configured to transition from the open-loop duty command to the closed-loop duty command in an event the comparison of the first and second comparators **152** and **153** indicate that the DC-link current is lower than the predetermined threshold value for a predetermined time and it is estimated whether, for the commanded motor speed and a load applied on the system, a closed-loop operation is possible without crossing the predetermined threshold value for the DC-link current. Based on an output of this estimation, the hoist application system **101** can transition to the closed-loop operation smoothly. In accordance with embodiments, the estimation can be carried out at a relatively slow rate so as to avoid unwanted switching between open-loop control to closed-loop control, which can result in system oscillation.

An operation of the hoist application system **101** is illustrated in FIG. **2**. As shown in FIG. **2**, a commanded motor speed can be set at 16,500 rpm. In this case, the motor is driven by the closed-loop duty command to 16,500 rpm, at which time a load on the motor is increased. This load increase continues until DC current reaches 150 A. At this point, the open-loop duty command is used to drive the motor with the commanded (or reference) speed dropped to 10,000 rpm and then to 4,000 rpm without load reduction. During the use of the open-loop duty command, it is observed that the measured motor speed more closely follows the commanded motor speed than in the moments following the load increase and until the transition to the open-loop control.

With reference to FIGS. **3** and **4**, a method **300** (see FIG. **3**) of operating a hoist application system, such as the hoist application system **101** of FIG. **1**, is provided. The method **300** includes driving a motor in accordance with a final duty command (block **301**), generating, in a closed-loop control unit, a closed-loop duty command (block **302**) by calculating the closed-loop duty command based on the error between the commanded motor speed and the measured motor speed (block **3021**), generating, in an open-loop control unit, an open-loop duty command (block **303**) and generating the final duty command from one of the closed-loop duty command and the open-loop duty command based on DC system information (block **304**). The generating of the open-loop duty command of block **303** can be executed using a fuzzy map algorithm as described above (block **3031**) or to achieve a gradual time-based duty cycle reduction (block **3032**).

The generating of the final duty command of block **304** can includes switching from the closed-loop duty command to the open-loop duty command (block **305**) or switching from the open-loop duty command to the closed-loop duty command (block **306**). The switching from the closed-loop duty command to the open-loop duty command of block **305** includes comparing the DC-link current to the predetermined threshold value (i.e., about 150 A) from the DC system information (block **3051**) and transitioning from the closed-loop duty command to the open-loop duty command in an event the DC-link current exceeds the predetermined threshold value for the predetermine time (block **3052**).

The switching from the open-loop duty command to the closed-loop duty command of block **306** can include determining whether the DC system information indicates that the DC-link current is lower than the predetermined threshold value for the predetermined time and estimating whether, for the commanded motor speed and a load applied on the system, a closed-loop operation is possible without crossing the predetermined threshold value for the DC-link current (block **3061**). In accordance with embodiments, the estimation can be carried out at a relatively slow rate to avoid unwanted switching between open-loop control to closed-loop control, which can result in system oscillation and, based on an output of this estimation, the hoist application system **101** can transition to the closed-loop operation smoothly. That is, the switching from the open-loop duty command to the closed-loop duty command of block **306** can further include switching or transitioning from the open-loop duty command to the closed-loop duty command in an event the DC system information indicates that the DC-link current is lower than the predetermined threshold value for the DC-link current for the predetermined time and, for the commanded motor speed and a load applied on the system, it is estimated that the closed-loop operation is possible without crossing the predetermined threshold value for the DC-link current (block **3062**).

As shown in FIG. **4**, a state diagram **401** illustrating a hoist application system operation is provided. The state diagram begins with closed-loop operation as described above. During the closed-loop operation, it is determined whether the DC current exceeds a threshold. If so, open-loop operation can be initiated as described above with the closed-loop duty cycle provided as an input and the open-loop duty cycle given as the output. During the open-loop operation, it is determined whether to return to closed-loop operation as explained above, or continue with the open-loop operation.

Technical effects and benefits of the present disclosure are the provision of a hoist application system that is low cost and has a relatively simple control algorithm.

The corresponding structures, materials, acts, and equivalents of all means or step-plus function elements in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed. The description of the present disclosure has been presented for purposes of illustration and description, but is not intended to be exhaustive or limited to the technical concepts in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the disclosure. The embodiments were chosen and described in order to best explain the principles of the disclosure and the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various embodiments with various modifications as are suited to the particular use contemplated.

While the preferred embodiments to the disclosure have been described, it will be understood that those skilled in the art, both now and in the future, may make various improvements and enhancements which fall within the scope of the claims which follow. These claims should be construed to maintain the proper protection for the disclosure first described.

What is claimed is:

1. A hoist application system, comprising:
a motor drive unit, which is receptive of a final duty command and which drives a motor in accordance with the final duty command;
a closed-loop control unit configured to generate a closed-loop duty command;
an open-loop control unit configured to generate an open-loop duty command; and
a switch logic unit, which is receptive of direct current (DC) system information, the closed-loop duty command and the open-loop duty command, and which is configured to generate the final duty command from one of the closed-loop duty command and the open-loop duty command based on the DC system information,
wherein:
the switch logic unit transitions from the closed-loop duty command to the open-loop duty command when a DC-link current exceeds a threshold value from the DC system for a predetermined time, and
the switch logic unit transitions from the open-loop duty command to the closed-loop duty command when the DC-link current is lower than the threshold value for the predetermined time and closed-loop operation is possible without crossing the threshold value.

2. The hoist application system according to claim **1**, wherein the closed-loop control unit is configured to calculate the closed-loop duty command based on an error between a commanded motor speed and a measured motor speed.

3. The hoist application system according to claim **1**, wherein the open-loop control unit is configured to calculate the open-loop duty command using a fuzzy map algorithm.

4. The hoist application system according to claim **3**, wherein the fuzzy map algorithm considers a measured motor speed, the DC system information and a current version of the final duty command.

5. The hoist application system according to claim **1**, wherein the open-loop control unit is configured to calculate the open-loop duty command to achieve a gradual time-based duty cycle reduction.

6. The hoist application system according to claim **1**, wherein the threshold value is about 150 A.

7. The hoist application system according to claim **1**, wherein the threshold value is 150 A.

8. A method of operating a hoist application system, the method comprising:
driving a motor in accordance with a final duty command;
generating, in a closed-loop control unit, a closed-loop duty command;
generating, in an open-loop control unit, an open-loop duty command; and
generating the final duty command from one of the closed-loop duty command and the open-loop duty command based on direct current (DC) system information,
wherein the generating of the final duty command comprises:
transitioning from the closed-loop duty command to the open-loop duty command when a DC-link current exceeds a threshold value from the DC system information for a predetermined time; and
transitioning from the open-loop duty command to the closed-loop duty command when the DC-link current is lower than the threshold value for the predetermined time and, for a commanded motor speed and a load applied on the system, the closed-loop operation is possible without crossing the threshold value.

9. The method according to claim **8**, wherein the generating of the closed-loop duty command comprises calculating the closed-loop duty command based on an error between a commanded motor speed and a measured motor speed.

10. The method according to claim **8**, wherein the generating of the open-loop duty command comprises calculating the open-loop duty command using a fuzzy map algorithm.

11. The method according to claim **10**, wherein the fuzzy map algorithm considers a measured motor speed, the DC system information and a current version of the final duty command.

12. The method according to claim **8**, wherein the generating of the open-loop duty command comprises calculating the open-loop duty command to achieve a gradual time-based duty cycle reduction.

13. The method according to claim **8**, wherein the threshold value is about 150 A.

14. The method according to claim **8**, wherein the threshold value is 150 A.

15. A method of operating a hoist application system, the method comprising:
driving a motor in accordance with a final duty command;
generating, in a closed-loop control unit, a closed-loop duty command;
generating, in an open-loop control unit, an open-loop duty command; and
generating the final duty command from one of the closed-loop duty command and the open-loop duty command based on direct current (DC) system information,
wherein the generating of the final duty command comprises one of switching from the closed-loop duty command to the open-loop duty command and switching from the open-loop duty command to the closed-loop duty command,
wherein:
the switching from the closed-loop duty command to the open-loop duty command comprises comparing a DC-link current to a threshold value from the DC system information and transitioning from the closed-loop duty command to the open-loop duty command in an event the DC-link current exceeds the threshold value for a predetermine time, and
the switching from the open-loop duty command to the closed-loop duty command comprises determining whether the DC system information indicates that a DC-link current is lower than a threshold value for a predetermined time and a closed-loop operation is possible without crossing the threshold value and switching from the open-loop duty command to the closed-loop duty command in an event the DC system information indicates that the DC-link current is lower than the threshold value for the predetermined time and, for a commanded motor speed and a load applied on the system, the closed-loop operation is possible without crossing the threshold value.

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